A SMOOTH MORPHISM IS COVERED BY SMOOTH MORPHISMS OF AFFINE SCHEMES WITH SECTION AND CONNECTED FIBRES

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 10956-10963 of sources/stacks-more-morphisms/pinned-original-source.tex; proof span is lines 10965-10994, the complete author proof (28 body lines).

Editorial note. Navigation stubs: 5 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Smooth morphisms, etale neighbourhoods of the base, and the behaviour of geometric connectedness of fibres under smooth morphisms with a section; the construction is retained.

Source reference. Original lemma lemma-connected-along-section-locally-constructible: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 8268–8273. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma lemma-connected-along-section-open: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 8387–8394. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma lemma-slice-smooth: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 10775–10783. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma section-connected-components: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 8172–8184. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 056U: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 10994–10994. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Date: stacks-project:more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Lemma 0.1. *Let $f : X \to S$ be a smooth morphism of schemes. Then there exists an étale covering $\{U_i \to X\}_{i \in I}$ such that $U_i \to S$ factors as $U_i \to V_i \to S$ where $V_i \to S$ is étale and $U_i \to V_i$ is a smooth morphism of affine schemes, which has a section, and has geometrically connected fibres.*

Proof. Let $s \in S$. By Varieties, Lemma 056U the set of closed points $x \in X_s$ such that $\kappa(x)/\kappa(s)$ is separable is dense in X_s. Thus it suffices to construct an étale morphism $U \to X$ with x in the image such that $U \to S$ factors in the manner described in the lemma. To do this, choose an immersion $Z \to X$ passing through x such that $Z \to S$ is étale (Lemma lemma-slice-smooth). After replacing S by Z and X by $Z \times_S X$ we see that we may assume $X \to S$ has a section $\sigma : S \to X$ with $\sigma(s) = x$. Then we can first replace S by an affine open neighbourhood of s and next replace X by an affine open neighbourhood of x. Then finally, we consider the subset $X^0 \subset X$ of Section section-connected-components. By Lemmas lemma-connected-along-section-open and lemma-connected-along-section-locally-constructible this is a retrocompact open subscheme containing σ such that the fibres $X^0 \to S$ are geometrically connected. If X^0 is not affine, then we choose an affine open $U \subset X^0$ containing x. Since $X^0 \to S$ is smooth, the image of U is open. Choose an affine open neighbourhood $V \subset S$ of s contained in $\sigma^{-1}(U)$ and in the image of $U \to S$. Finally, the reader sees that $U \cap f^{-1}(V) \to V$ has all the desired properties. For example $U \cap f^{-1}(V)$ is equal to $U \times_S V$ is affine as a fibre product of affine schemes. Also, the geometric fibres of $U \cap f^{-1}(V) \to V$ are nonempty open subschemes of the irreducible fibres of $X^0 \to S$ and hence connected. Some details omitted. $\square$